Scalability & Future Plan

Date	6 July 2026
Team ID	XXXXXX
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	1 Mark

Scalability & Future Plan:
The Flood Prediction using Machine Learning system has been designed with a modular architecture that allows future enhancements and expansion. Although the current implementation is intended for demonstration and small-scale deployment, the application can be scaled to support larger datasets, multiple users, and real-time prediction services. Future improvements include cloud deployment, integration with live weather APIs and IoT-based flood monitoring sensors, enhanced security mechanisms, and the adoption of advanced machine learning models to improve prediction accuracy and system reliability.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	The application is deployed on a local Flask server and supports only limited concurrent users.	Reduces accessibility and scalability for public usage.	High
2	Predictions are based on historical dataset values without real-time weather data integration.	Prediction accuracy may decrease under rapidly changing environmental conditions.	High
3	The current system stores limited data locally without centralized database support.	Difficult to manage large datasets and historical prediction records.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a small number of local users.	Deploy the application on cloud platforms such as AWS, Azure, or Google Cloud with load balancing to support thousands of users.
2	Data Storage	Uses local files for dataset storage.	Integrate a scalable database such as MySQL, PostgreSQL, or MongoDB to manage large volumes of data efficiently.
3	Performance	Prediction is executed on a single local server.	Optimize the machine learning model, implement caching, and use cloud computing resources for faster response times.
4	Security	Basic application security is implemented.	Add user authentication, HTTPS encryption, secure APIs, role-based access control, and regular security monitoring.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Deploy the application on a cloud hosting platform with database integration.	1–2 Months	Improved accessibility, availability, and scalability.
Phase 3	Integrate live weather APIs and river-level monitoring services for real-time prediction.	2–4 Months	More accurate and timely flood predictions.
Phase 4	Connect IoT sensors for continuous environmental data collection and automated monitoring.	4–6 Months	Real-time flood detection with minimal manual intervention.